Ask or Recommend: An Empirical Study on Conversational Product Search

Heli Ma
University of Electronic Science and Technology of China
Chengdu, China
1357137816@qq.com

Jie Zou*
University of Electronic Science and Technology of China
Chengdu, China
zoujie0806@gmail.com

Mohammad Aliannejadi
University of Amsterdam
Amsterdam, The Netherlands
m.aliannejadi@uva.nl

Evangelos Kanoulas
University of Amsterdam
Amsterdam, The Netherlands
E.Kanoulas@uva.nl

Yi Bin
Tongji University
Shanghai, China
yi.bin@hotmail.com

Yang Yang
University of Electronic Science and Technology of China
Chengdu, China
yang.yang@uestc.edu.cn

Abstract

Conversational Product Search (CPS) provides an engaging way for users to find products through effective natural language conversations. However, understanding the effect of conversational characteristics on user search performance and when to ask clarifying questions or recommend products remains unexplored. To fill the gap, we conduct an empirical study in this paper. Specifically, we developed a conversational system that allows participants to join as customers or shopping assistants, to simulate the conversational product search activity. Data collected from conversations and participant feedback indicate that: (a) CPS systems tend to ask clarifying questions early in the conversation when users express the intent of issuing a new query and chitchat, while they tend to recommend products at a later stage of conversations; asking clarifying questions early and recommending products lately can significantly improve search performance and user's satisfaction; (b) asking clarifying questions and more fine-grained search keywords positively influence search performance in terms of finding relevant products; (c) although the conversation time has a positive impact on the number of recommended products, the performance gain diminishes with longer conversation time; (d) more clarifying questions, more conversation turns, and longer system response time lead to decreased user satisfaction.

CCS Concepts

• **Information systems → Users and interactive retrieval**; • **Human-centered computing → Empirical studies in HCI**.

Keywords

Empirical Study; Product Search; Conversational Product Search.

ACM Reference Format:
Heli Ma, Jie Zou, Mohammad Aliannejadi, Evangelos Kanoulas, Yi Bin, and Yang Yang. 2024. Ask or Recommend: An Empirical Study on Conversational Product Search. In *Proceedings of the 33rd ACM International Conference on Information and Knowledge Management (CIKM '24), October 21–25, 2024, Boise, ID, USA.* ACM, New York, NY, USA, 5 pages. https://doi.org/10.1145/3627673.3679875

1 Introduction

Traditional product search systems typically require users to input queries and browse search results to locate their desired products [1, 13, 17]. However, this approach is often inefficient and may lead to mismatches between search results and user needs, due to semantic gaps between queries and products[7]. To overcome these challenges, CPS systems enable natural language interactions with users and collect explicit feedback to better understand user information or his/her needs [16, 20].

Zhang et al. [14] proposed a first CPS model that incorporates a multi-memory network architecture. Similarly, Bi et al. [3] proposed a CPS model with a focus on negative feedback. Zou and Kanoulas [17] introduced a sequential Bayesian method that employs cross-user duet training to enhance CPS. Most recently, Zou et al. [16] proposed a CPS model based on representation learning. The aforementioned studies on CPS demonstrate the effectiveness of CPS, however, they focus on the development of CPS algorithms and they do not explore the potential mechanism of CPS from users' perspectives. There are only a few existing research focusing on user studies on CPS. Papenmeier et al. [8] conducted an empirical study to analyze strategies that experts and customers employ to communicate about products in CPS. Building on this, Papenmeier and Topp [9] compared chatbot behaviors with different types of feedback in CPS. Meanwhile, Javadi et al. [5] explored subjective narratives in conversations in CPS. In a related vein, Zou et al. [18] conducted an empirical study on clarifying question-based product search systems, shedding light on the existing question-based product search approaches. However, to the best of our knowledge, the extent to which conversational characteristics affect user search

*Corresponding author.

CIKM '24, October 21–25, 2024, Boise, ID, USA

ACM ISBN 979-8-4007-0436-9/24/10
https://doi.org/10.1145/3627673.3679875

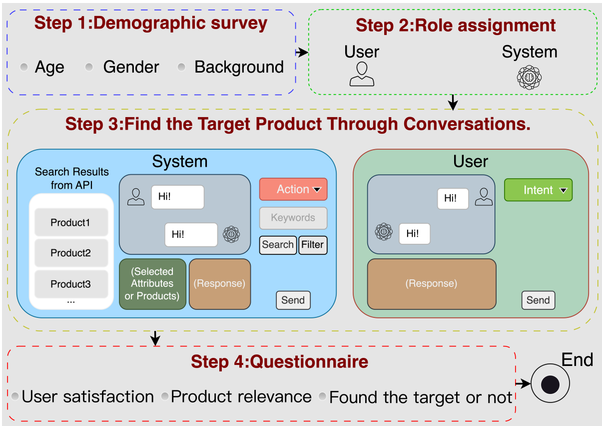


Figure 1: System framework

performance and when to ask clarifying questions or recommend products remains unexplored.

In this study, we conduct an empirical study to bridge this research gap. We developed an experimental system that involved online participants, with some participants mimicking the roles of online shopping assistants (the CPS system), while others acted as customers. We collected data on conversation, explicit feedback, and implicit behaviors from participants, aiming to address the following three research questions:

(1) **RQ1:** When should the system pose clarifying questions or recommend products to the users?
(2) **RQ2:** What are the factors that affect user search performance in terms of finding relevant information in CPS?
(3) **RQ3:** How do different factors influence user satisfaction in CPS?

We believe that answering these research questions can offer valuable insights for the evaluation and design of future CPS systems.

2 Method

2.1 Study Design

In this study, we developed an online Web system in a wizard-of-oz fashion, where some participants act as customers, while others as shopping assistants. Customers express their need for products and obtain their target products through conversations with the shopping assistants. The framework of our system is shown in Figure 1. All included participants go through four steps.

Step 1: Demographic Survey. First, participants are asked to complete a demographic survey, filling in their gender, age, and career field.

Step 2: Role Assignment. In this step, we randomly assign a role, either the customer or shopping assistant, to the participant, following the related literature [4, 15] assigning conditions or subjects randomly. Participants mimicking the customers are asked to imagine target products they want to purchase in mind, and then initiate a conversation with the assistant to obtain the target products with the assistant's help.

Step 3: Find the Target Product Through Conversations. The customers chat with the assistants through conversations in the chat room to find their target products. Before conversing with the assistant, they choose the user 'intent' of their message from a predefined list. The assistant can then respond by specifying the kind of system 'action' they like. The user intent and system action are designed to enhance the analysis of the customer and assistant's behavior during the conversation, where the user intent and system action taxonomies are based on previous conversational systems [2, 11]. The user intents include 'Reveal', 'Revise', 'Interpret', and 'Chitchat'. 'Reveal' indicates that customers intend to disclose their requirements for target products to the assistants, 'Revise' is for revising their product requirements, 'Interpret' means responding to assistants' clarifying questions or feedback on recommendations, and 'Chitchat' is for greetings.

The system actions include 'Chitchat', 'Clarify', 'Recommend', and 'No-answer'. 'Chitchat' is for greeting, usually in response to a customer's greeting. 'Clarify' means asking clarifying questions [10] to clarify customers' product needs. 'Recommend' means suggesting products to customers. 'No-answer' means the product is not available in the repository and they can not find related products to meet customers' needs. Assistants only recommend products after they select 'Recommend' action. Additionally, a dedicated search engine built on the Amazon product search API enables assistants to submit keyword queries and retrieve detailed lists of recommended products from Amazon (By the 'search' button in Figure 1).[1] That is, during the conversation, the assistant can ask clarifying questions or select products to recommend to the customer based on the search results. The conversation can be multiple turns and customers can end the conversation at any time.

Step 4: Questionnaire Survey. In this step, customers are asked to respond to questions regarding their experience with the "system", in terms of the rating of satisfaction (scale of 1 to 5). Also, there is a relevance judgment page, with the customer evaluating the relevance of each product recommended by the assistant in the conversations.

2.2 Participants and Quality Control

Participants in the experiment were invited via email, with 196 individuals (students and staff from two universities in Europe and Asia) participating in the study. Participants received an average compensation of approximately 2.5 GBP for each conversation.

To ensure data quality, we performed four quality checks: (1) all participants received on-site training with detailed video and document references; (2) we evaluated the time participants spent reading the textual descriptions and asked participants questions about the study to ensure that they have read and understood the instructions; (3) we exclude the data with incomplete conversations or incomplete item relevance judgments; (4) we manually check the collected data and correct typos, grammatical errors, and wrong labels regarding intents and actions.

3 Results

The data comprises 904 conversations, 5,254 utterances, and 2,491 recommended products. Each conversation contains 5 to 16 utterances, 3 to 7 turns, and 1 to 30 products. The data statistics are reported in Table 1. Unless otherwise reported, we perform t-tests

[1] https://www.amazon.com/

Table 1: Statistics of the collected data.

# conversations	904	# participants	196
# clarifying questions	291	# search requests	1700
# utterances	5,254	avg. # utterances	5.81
# turns	3,044	avg. # turns	3.37
# items	2,491	avg. # items	2.76

[6] for comparisons between two groups only, and perform one-way ANOVA and Tukey's HSD tests for comparisons with more than two groups, following previous user studies [12].

3.1 RQ1: When to Ask & When to Recommend

In the analysis that follows we try to understand what factors relate to the system (assistant) asking a clarifying question or making product recommendations to the user (customer).

3.1.1 *Clarifying Questions.*

User Intent. We first explore whether the user intent proceeds with the assistant's decision to ask clarifying questions. From the analysis, we observe that, when customers select the 'Chitchat' and 'Reveal' intent, the assistant poses clarifying questions 74.8% and 9.68% of the time, respectively. In contrast, for 'Interpret' and 'Revise' intents, clarifying questions are posed in less than 1% of the instances. The one-way ANOVA test reveals significant differences among different intents (p<0.05). This indicates that the assistant tends to ask clarifying questions when users select 'Chitchat' and 'Reveal' intents, which typically occur in the early stages of dialogues. Further, we explore the impact of asking clarifying questions on search performance and user satisfaction when customers select different intents. There are significant decreases in the proportion of found target products (95.19%, 87.50%, and 66.67% on average for 'Chitchat', 'Reveal', and 'Interpret', respectively) and user satisfaction (4.84, 4.59, and 4.17 on average for 'Chitchat', 'Reveal', and 'Interpret', respectively) from the early stage of 'Chitchat' and 'Reveal' to the later stage of 'Interpret' (significant differences, p<0.05). This confirms that the system should ask clarifying questions during the early stages of intent. When customers select 'Chitchat' for greeting at the beginning of conversations, the assistant often seeks clarification due to a lack of information. After customers select 'Reveal' to explain their needs, fewer clarifying questions are needed for them. However, ambiguous or challenging requests may still trigger specific clarification. 'Interpret' and 'Revise' actions typically occur later in the conversation when enough information is available and thus clarifying questions are seldom required.

SERP Diversity. We explore whether the diversity of the Search Engine Results Page (SERP) affects the necessity of asking clarifying questions, which is shown in Figure 2 (left). We use nERR-IA@20 as the diversity metric. We observe that, with higher diversity (higher score of nERR-IA@20), the assistant tends to ask more clarifying questions. This is natural since higher diversity indicates the availability of many options that fulfill the declared need of the customer and thus need to be clarified. This is in line with Zou et al. [15] that customers engage more with clarifying questions as diversity increases. The highly diverse results lead to the assistant being inspired to ask more questions to more accurately locate the target products.

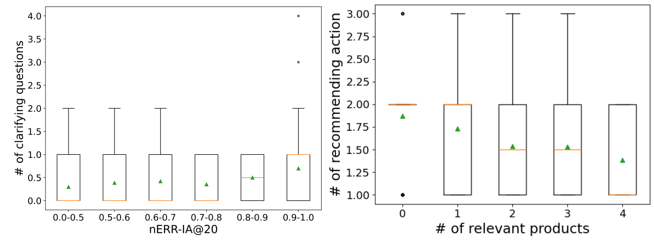


Figure 2: The effects of SERP diversity and number of relevant products on when to ask/recommend.

3.1.2 *Product Recommendation.*

User Intent. We observe the assistant recommends products 3.5% of the time when users express the 'Chitchat' intent. In contrast, recommendations occur 86.3% of the time for the 'Reveal' intent, 88.9% for 'Interpret', and 97.1% for 'Revise'. The one-way ANOVA test confirms the significance of these differences (p<0.05). This suggests that product recommendations often occur in the later stages of dialogues, following user intents like 'Interpret', and 'Revise'. Additionally, we observe there are significant increases in the proportion of found target products (90.59%, 97.81%, and 100% on average for 'Reveal', 'Interpret' and 'Revise', respectively) and user satisfaction (4.49, 4.96, and 4.86 on average for 'Reveal', 'Interpret' and 'Revise', respectively) from the early stage of 'Reveal' to the later stage of 'Interpret' and 'Revise' (significant differences, p<0.05). This suggests the system should make recommendations during the later stages of Intent. After the later stage intents, customers convey effective information to the assistant, and thus the assistant can identify relevant products for the customer, which is consistent with the conclusion of Javadi et al. [5].

Relevant Products. In addition, we explore whether the number of found relevant products affects the assistant's recommendation behavior. The results, as shown in Figure 2 (right), demonstrate that with the increase in the number of relevant products in a conversation, the number of the assistant's recommendation behavior decreases. This is, with an increase in the number of relevant products, recommending the right products to customers becomes easier, which reduces the need to recommend new ones. Conversely, when relevant products are few, the assistant tends to employ multiple recommendation actions to gradually locate the target products.

3.1.3 *Summary.* The system prefers to ask clarifying questions at the beginning of the conversation and recommend products in the later stages, which will significantly improve search performance and user satisfaction. Meanwhile, higher product diversity leads to asking more clarifying questions. Assistants tend to recommend more times when there are few numbers of relevant products.

3.2 RQ2: Factors Improving Search Performance

Search Keywords. We observe that the increase in the number of search keywords in one conversation round, which are more fine-grained searches, leads to more recommended and relevant products (# of recommended products: 2.57, 4.16, and 7.12 on average for 1, 2, and 3 keywords respectively; # of relevant products: 1.24, 2.17, and 3.41 on average for 1, 2, and 3 keywords respectively). The results suggest that, for complex and ambiguous user needs, the assistant needs to use multiple and fine-grained keywords for

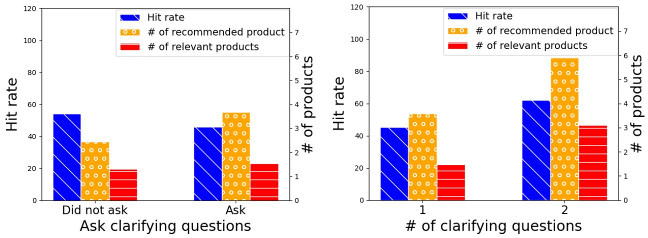


Figure 3: The effects of clarifying questions on search performance.

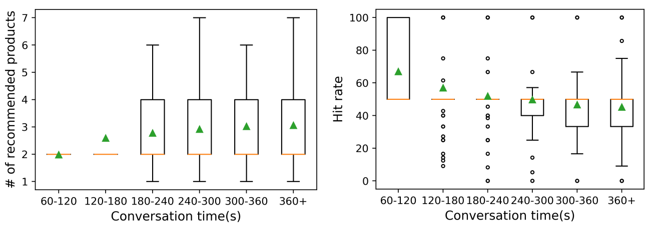


Figure 4: The effects of conversation time on search performance.

search, leading to more accurate search results. It inspires the assistant to recommend more products and also brings more relevant products.

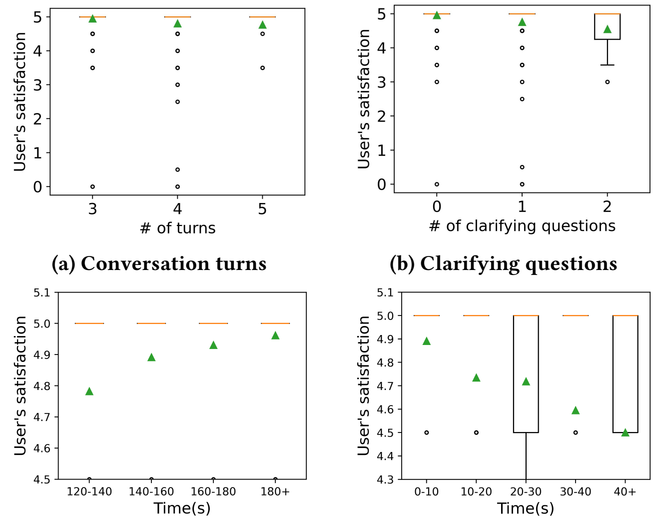
Clarifying Questions. Observe from Figure 3, asking clarifying questions leads to more recommended and related products. We also observed the same trend when asking more clarifying questions. This is in line with that the performance of CPS models increases with the increase in the number of clarifying questions [16]. This suggests that, for some customers, their target products are simple, clear, and explicit, and their initial expression of intent enables the system to find the target products without the need to ask clarifying questions. However, when faced with vague and indirect user needs, the system often needs to ask clarifying questions to effectively locate the target products.

Conversation Time. Figure 4 (right) shows that longer conversations correlate with a lower hit rate of recommended products (i.e., the ratio of relevant products to recommended products). This suggests that while more time increases the number of recommendations (Figure 4 (left)), it does not necessarily benefit customers. Our data analysis indicates that conversation time is significantly affected by the complexity of customer demands. As understanding customer needs becomes more difficult, conversation time increases, and the relevance of recommended products declines.

Summary. Fine-grained keywords and clarifying questions are beneficial to the system's performance, and an increase in conversation time does not necessarily yield additional benefits.

3.3 RQ3: Factors Affecting the User Satisfaction

Conversation Turns. Figure 5a illustrates an increase in the number of turns leads to a significant decline in user satisfaction ratings (p<0.05). This is in line with that customers expect the assistant can recommend high-quality products with fewer conversation rounds in real applications.



(a) Conversation turns **(b) Clarifying questions**

(c) Reveal time **(d) Clarify time**

Figure 5: The effects of # conversation turns, # clarifying questions, the time used by 'Reveal' intent, and time spent on 'Clarify' action, on user's satisfaction.

Clarifying Questions. Figure 5b shows more questions lead to a significant decrease in user satisfaction (p<0.05). Increased question numbers, in a sense, reflect an extra burden for customers to provide feedback, leading to a negative impact on user satisfaction [19].

Response Time. In Figure 5c, we observe that longer customer 'Reveal' time (time taken for selecting 'Reveal' intent and generating a response) leads to increased satisfaction. This might be because more 'Reveal' time leads to a more accurate expression of customer demands for products, which is beneficial for locating the target product successfully and thus increasing satisfaction. Moreover, Figure 5d shows that the longer the assistant takes to generate clarifying questions, the lower the user satisfaction. This highlights user dissatisfaction with waiting for assistants' responses.

Summary. More conversation turns, more clarifying questions, and more time spent generating clarifying questions lead to user dissatisfaction. However, more time spent by customers on the 'Reveal' intent results in higher user satisfaction.

4 CONCLUSION AND DISCUSSION

In this paper, we conduct an empirical study to explore when to ask clarifying questions and recommend products, as well as different factors affecting user search performance and satisfaction. To this end, we deploy an online CPS system and collect participants' interaction data and questionnaires for analysis. We found that early-stage intents in the conversation and the high diversity of search results lead to the necessity of asking clarifying questions for clarification, while later-stage intents lead to recommending products, which usually means high confidence and enough information has been gathered. It is interesting that spending more time in a conversation does not necessarily result in better search performance and user satisfaction. Users usually expect a quick-responding system; Although more clarifying questions and a longer conversation time lead to a greater number of recommended products, it may raise the risk of user dissatisfaction.

References

[1] Qingyao Ai, Yongfeng Zhang, Keping Bi, Xu Chen, and W Bruce Croft. 2017. Learning a hierarchical embedding model for personalized product search. In *Proceedings of the 40th International ACM SIGIR Conference on Research and Development in Information Retrieval*. 645–654.

[2] Leif Azzopardi, Mateusz Dubiel, Martin Halvey, and Jeffery Dalton. 2018. Conceptualizing agent-human interactions during the conversational search process. In *The second international workshop on conversational approaches to information retrieval*.

[3] Keping Bi, Qingyao Ai, Yongfeng Zhang, and W Bruce Croft. 2019. Conversational product search based on negative feedback. In *Proceedings of the 28th acm international conference on information and knowledge management*. 359–368.

[4] Morgan Harvey and Matthew Pointon. 2017. Searching on the go: the effects of fragmented attention on mobile web search tasks. In *Proceedings of the 40th International ACM SIGIR Conference on Research and Development in Information Retrieval*. 155–164.

[5] Vahid Sadiri Javadi, Martin Potthast, and Lucie Flek. 2023. OpinionConv: Conversational Product Search with Grounded Opinions. In *Proceedings of the 24rd Annual Meeting of the Special Interest Group on Discourse and Dialogue*.

[6] Tae Kyun Kim. 2015. T test as a parametric statistic. *Korean journal of anesthesiology* 68, 6 (2015), 540–546.

[7] Hang Li, Jun Xu, et al. 2014. Semantic matching in search. *Foundations and Trends® in Information Retrieval* 7, 5 (2014), 343–469.

[8] Andrea Papenmeier, Alexander Frummet, and Dagmar Kern. 2022. ‚ÄúMhm...‚Äù– Conversational Strategies For Product Search Assistants. In *Proceedings of the 2022 Conference on Human Information Interaction and Retrieval*. 36–46.

[9] Andrea Papenmeier and Elin Anna Topp. 2023. Ah, Alright, Okay! Communicating Understanding in Conversational Product Search. In *Proceedings of the 5th International Conference on Conversational User Interfaces*. 1–5.

[10] Sudha Rao and Hal Daumé. 2018. Learning to Ask Good Questions: Ranking Clarification Questions using Neural Expected Value of Perfect Information. In *Proceedings of the 56th Annual Meeting of the Association for Computational Linguistics*. 2737–2746.

[11] Pengjie Ren, Zhongkun Liu, Xiaomeng Song, Hongtao Tian, Zhumin Chen, Zhaochun Ren, and Maarten de Rijke. 2021. Wizard of Search Engine: Access to Information Through Conversations with Search Engines. In *Proceedings of the 44th International ACM SIGIR Conference on research and development in information retrieval*. 533–543.

[12] Andrew Turpin and Falk Scholer. 2006. User performance versus precision measures for simple search tasks. In *Proceedings of the 29th annual international ACM SIGIR conference on Research and development in information retrieval*. 11–18.

[13] Christophe Van Gysel, Maarten de Rijke, and Evangelos Kanoulas. 2016. Learning Latent Vector Spaces for Product Search. In *Proceedings of the 25th ACM International on Conference on Information and Knowledge Management*. 165–174.

[14] Yongfeng Zhang, Xu Chen, Qingyao Ai, Liu Yang, and W Bruce Croft. 2018. Towards conversational search and recommendation: System ask, user respond. In *Proceedings of the 27th ACM International Conference on Information and Knowledge Management*. 177–186.

[15] Jie Zou, Mohammad Aliannejadi, Evangelos Kanoulas, Maria Soledad Pera, and Yiqun Liu. 2023. Users meet clarifying questions: Toward a better understanding of user interactions for search clarification. *ACM Transactions on Information Systems* 41, 1 (2023), 1–25.

[16] Jie Zou, Jimmy Huang, Zhaochun Ren, and Evangelos Kanoulas. 2022. Learning to ask: Conversational product search via representation learning. *ACM Transactions on Information Systems* 41, 2 (2022), 1–27.

[17] Jie Zou and Evangelos Kanoulas. 2019. Learning to ask: Question-based sequential Bayesian product search. In *Proceedings of the 28th ACM International Conference on Information and Knowledge Management*. 369–378.

[18] Jie Zou, Evangelos Kanoulas, and Yiqun Liu. 2020. An Empirical Study on Clarifying Question-Based Systems. In *Proceedings of the 29th ACM International Conference on Information & Knowledge Management*. 2361–2364.

[19] Jie Zou, Aixin Sun, Cheng Long, Mohammad Aliannejadi, and Evangelos Kanoulas. 2023. Asking Clarifying Questions: To benefit or to disturb users in Web search? *Information Processing & Management* 60, 2 (2023), 103176.

[20] Jie Zou, Aixin Sun, Cheng Long, and Evangelos Kanoulas. 2024. Knowledge-Enhanced Conversational Recommendation via Transformer-based Sequential Modelling. *ACM Trans. Inf. Syst.* (jul 2024).